

C150: an exactly solved quantum cellular automaton

...and exactly what the solution does not determine
Krylov sector analysis, Tier 1 — report R8

The rule, and one naming note

- **C150 = Wolfram 150 = local symbol table (I, V, V, I)**
 - Hadamard fires at site $i \iff x(i-1) \oplus x(i+1) = 1$
 - — the firing condition is exactly classical ECA-90's update
- **Boundary: obc0, the missing neighbour reads 0**
- **Cycle: brick-wall — even sublattice first, then odd**
- **Same object as HSF rule 6 with gate H (checked from HSF/src/model.jl)**
 - — so the whole Julia eigendata set is a usable independent oracle
- **Why this rule**
 - R2 saw binomial sector sizes; R1 uses it as the HSF regression fixture;
 - and its fitted growth exponent 2.022 sat above the informational ceiling 2

Five results

- **1. Reduction — sector partition = a single-flip graph**
 - for ANY unitary rule; the brick-wall ordering drops out
- **2. Solution — walls, hard-core hopping, closed form**
 - $|S_w| = C(N+1, w)$ for even w ; the charge is the Ising domain-wall number
- **3. C150 is symmetry-resolved, NOT fragmented**
 - sector count is linear in N — it is the family's null model
- **4. Interacting, but not chaotic**
 - #distinct eigenvalues = $3^{\lfloor N/2 \rfloor}$ exactly — no RMT ensemble fits
- **5. Area law vs volume law is a sector statement**
 - ...but the sector fixes only the ceiling, not the value

Result 1 — the sector partition is a flip graph

- **Theorem. For a unitary rule the sectors are the connected components of**
 - $x \sim x \oplus 2^i$ for every site i where the Hadamard fires
- **Proof sketch**
 - each half-layer is depth-1 $\Rightarrow$ its image from $|x\rangle$ is a subcube (R1 no-interference)
 - x lies in its own image ($\langle b|H|b\rangle \neq 0$) $\Rightarrow$ composite image $\supseteq$ both half-layer images
 - inside a half-layer the V/I pattern is frozen $\Rightarrow$ the subcube is single-flip connected
- **Two consequences**
 - the brick-wall ORDER is irrelevant — the partition is a property of the symbol table
 - cost drops from $O(2^N \cdot |\text{succ}|)$ to $O(N \cdot 2^N)$ — this is what reaches $N = 32$
- **Validated: all 16 unitary rules $\times$ both boundaries $\times N = 3 \dots 12$ — 320 units, 0 mismatches**

Result 2 — walls: the whole rule in one line

- **Bond variables** $b_j = x_j \oplus x_{j+1}$, $j = 0 \dots N$, $x(-1) = x(N) = 0$
 - $x \mapsto b$ is a BIJECTION onto the even-weight strings of length $N+1$
- **The elementary move**
 - flipping x_i toggles bonds b_i, b_{i+1} , and is allowed iff $b_i \neq b_{i+1}$
 - $\Rightarrow$ a HARD-CORE HOP of one domain wall. Walls are never created or destroyed.
- **Therefore**
 - conserved charge $Q = \sum (1 - Z_j Z_{j+1})/2$ — the open Ising domain-wall number
 - each wall shell is exactly one sector (hard-core hopping is connected on a shell)
- **Zero charge violations over every state, $N = 3 \dots 12$, both boundary conditions**

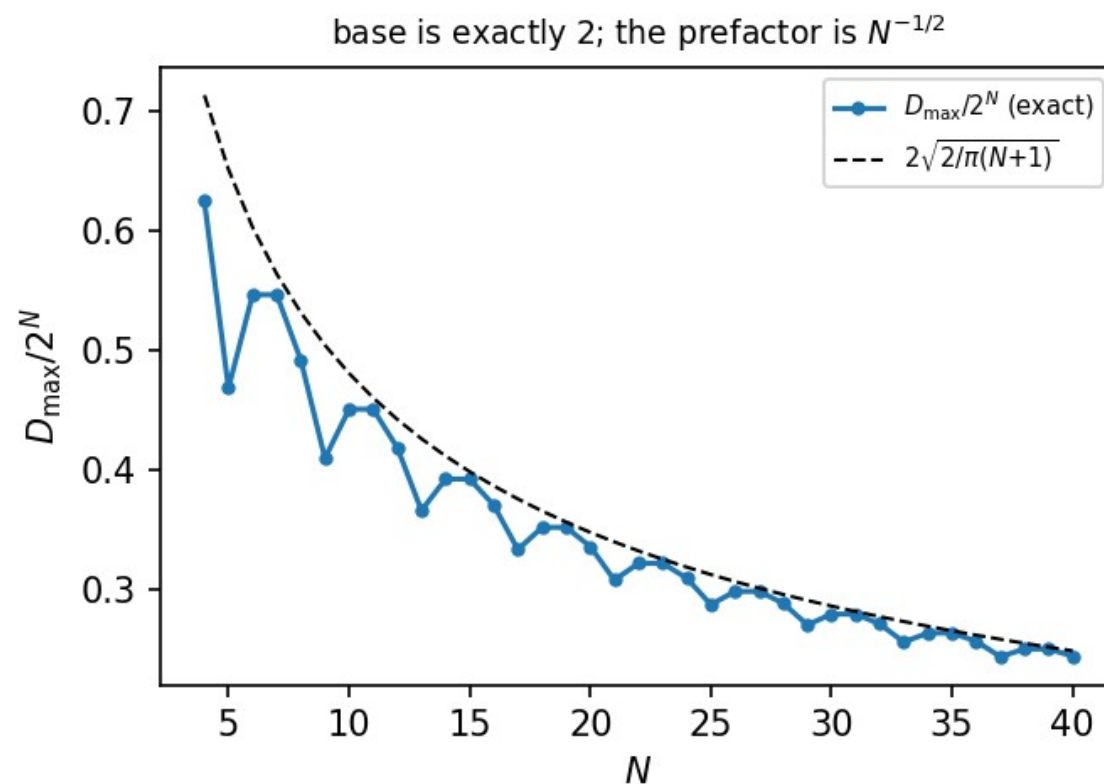
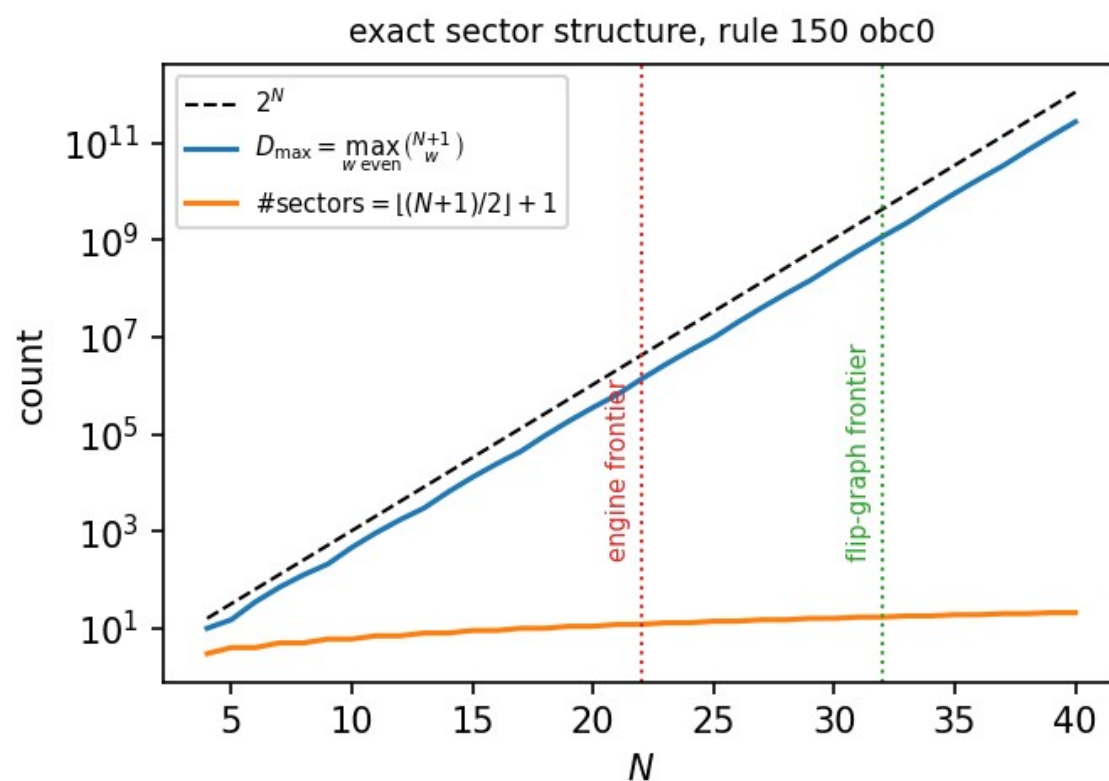
Result 3 — the closed form, and the growth law

- **$|S_w| = C(N+1, w)$ for even w **#sectors** = $\lfloor (N+1)/2 \rfloor + 1$ **$D_{\max}$ = max over even w**
 - matches every stored Tier-1a unit, $N = 6 \dots 20$, both boundary conditions**
- **C150 is NOT fragmented**
 - sector count is LINEAR in N and every sector is a full Q -eigenspace
 - no Krylov structure beyond the $U(1)$ charge — contrast W204's 2^N frozen singletons
 - $\rightarrow$ it is the wrong example to quote for Hilbert-space fragmentation
- **The 2.022 anomaly, resolved**
 - $D_{\max}/2^N = 2\sqrt{2/\pi(N+1)}$ — base is EXACTLY 2, the decay is an $N^{-1/2}$ prefactor
 - plus a sawtooth at $N \equiv 1 \pmod{4}$, where the central binomial sits at odd w
 - a pure-exponential fit reads prefactor + sawtooth as a base above the ceiling

Numerical frontier — two independent routes

N	2^N	#sectors	D_max	streamed engine	flip graph
20	1,048,576	11	352,716	1.1 h	—
21	2,097,152	12	646,646	3.6 h	0.3 s
22	4,194,304	12	1,352,078	10.2 h	0.4 s
28	268,435,456	15	77,558,760	—	37.7 s
30	1,073,741,824	16	300,540,195	—	154 s
31	2,147,483,648	17	601,080,390	—	297 s
32	4,294,967,296	17	1,166,803,110	—	815 s

Exact sector structure, and where the numerics stop

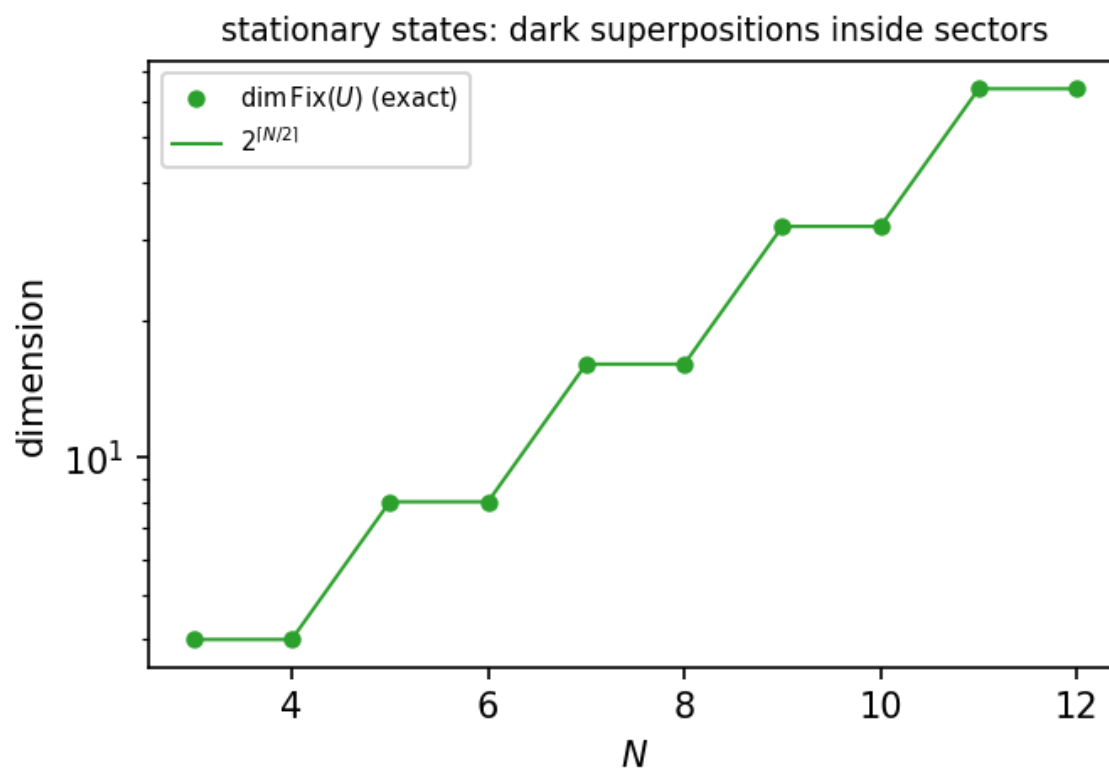
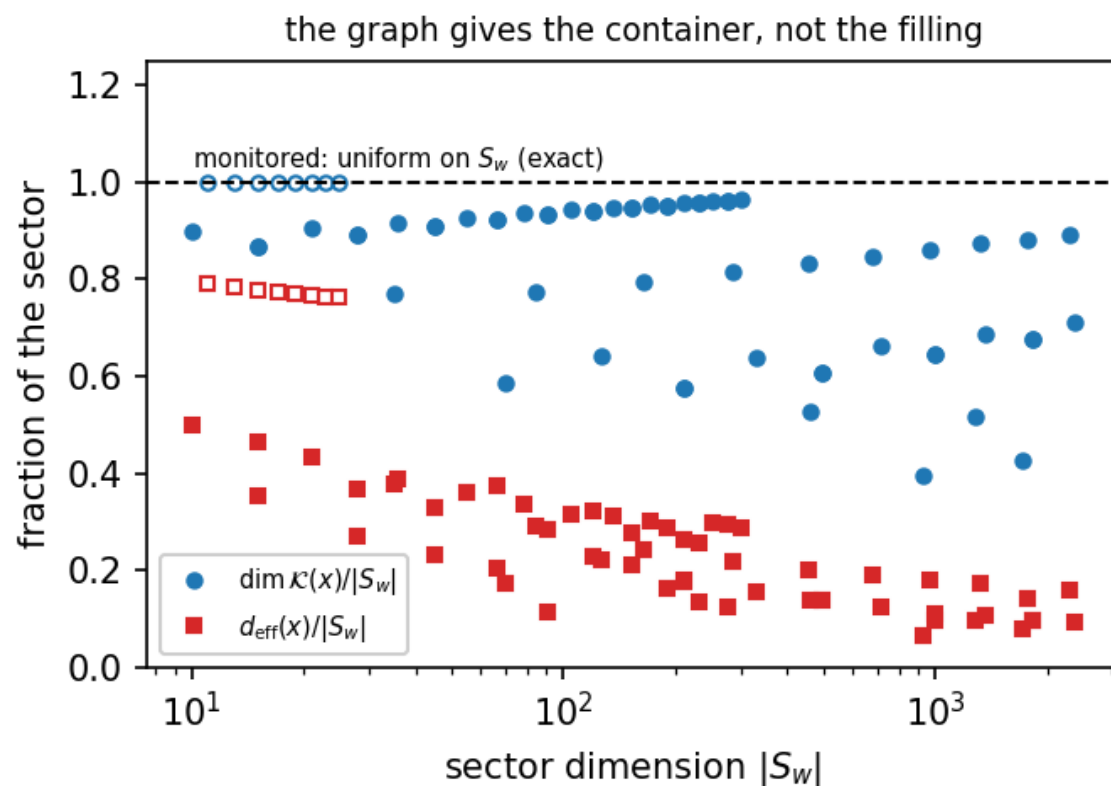


Left: the closed form with both computed frontiers marked. Right: $D_{\max}/2^N$ against $2\sqrt{2/\pi(N+1)}$ — base 2, $N^{-1/2}$ prefactor, sawtooth at $N \equiv 1 \pmod{4}$.

What the sector does NOT determine

- **A sector is the maximal reachable set from a basis state — nothing more**
- **An exponentially large stationary space**
 - $\dim \text{Fix}(U_w) = C([N/2], w/2)$, $\dim \text{Fix}(U) = 2^{[N/2]} = 2^{(\text{\#even sites})}$
 - $\text{Fix}(U) = \text{Fix}(L_A) \cap \text{Fix}(L_B)$; almost all of it is DARK SUPERPOSITIONS
- **Hence an inner sector is never a Krylov space**
 - spectrum degenerate $\iff |S_w| > N+1$; $\dim K(x)/|S_w| = 0.40 \dots 0.96$
- **Monitored vs unmonitored disagree inside a sector**
 - monitored: provably uniform on S_w (doubly stochastic, irreducible, aperiodic)
 - unmonitored: diagonal ensemble occupies only 6.6 % ... 50 % of the sector
- **Reflection: commutes for odd N ; conjugates U to U^{-1} for even N**

The graph gives the container, not the filling



Left: Krylov and diagonal-ensemble fractions of a sector; the dashed line is the monitored answer, which is exactly uniform. Right: $\dim \text{Fix}(U) = 2^{\lfloor N/2 \rfloor}$.

Result 4a — interacting, not free

- **In the bond basis the gate at site i touches bonds $i, i+1$ only**
 - identity when both bonds agree; on the one-wall subspace a Hadamard coin
 - $M(c) = (1/\sqrt{2})[[-(-1)^c, 1], [1, (-1)^c]]$, $\det M = -1$, c = wall parity to the left
 - this bond circuit reproduces the engine to 1.1×10^{-16} — it IS C150
- **Two diagonal signs spoil gaussianity**
 - $\det(H) = -1$ but the gate acts by $+1$ on a doubly occupied pair $\rightarrow$ maximal nearest-neighbour wall interaction $\exp(i\pi n n)$
 - the $(-1)^c$ string sits on the DENSITY term, not the hop $\rightarrow$ not absorbed by Jordan-Wigner
- **Sharp test: a Gaussian circuit has additive many-body eigenphases**
 - C150: 0.55 ... 1.87 rad remove one defect: still 0.43 ... 1.10 rad
 - remove BOTH: additive to 4×10^{-15} — so both signs are needed, and C150 is interacting

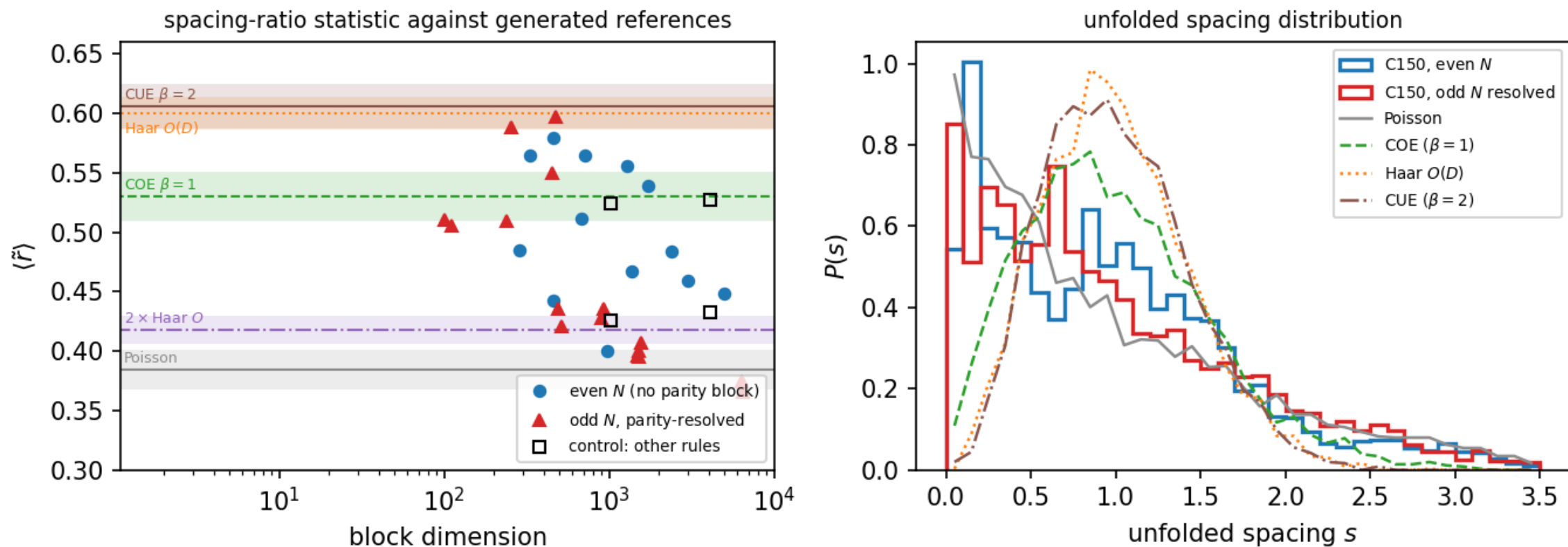
Result 4b — the spectrum is exponentially degenerate

N	2^N	#distinct	closed form $3^m / (3^m \pm 1)/2$	fraction
8	256	81	81	0.316
9	512	121	121	0.236
10	1,024	243	243	0.237
11	2,048	365	365	0.178
12	4,096	729	729	0.178
13	8,192	1,093	1,093	0.133
14	16,384	2,187	2,187	0.133

Result 4c — no random-matrix ensemble fits

- **$\beta = 2$ (GUE / CUE) excluded by symmetry**
 - U is real orthogonal $\Rightarrow$ complex conjugation is a time reversal, $K^2=1$
 - $\Rightarrow$ eigenvalues in mirror pairs; all statistics on $\theta \in (0, \pi)$ only
- **$\beta = 1$ (COE / Haar-O) excluded by the degeneracy law**
 - no chaotic Floquet operator has 3^m distinct eigenvalues out of 2^N
- **Poisson excluded too**
 - ONE universal set Θ_N per N: every shell's spectrum $\subseteq \Theta_N$, and the largest shell realises ALL of it (small shells as little as 1.8 %)
 - largest shell $N=16$, $w=8$: $\langle \tilde{r} \rangle = 0.347$, BELOW Poisson (0.384) — and superposition can only approach Poisson from above
- **Structural source: $U = L_B \cdot L_A$ is a product of two involutions**
 - gives the $\pm\theta$ pairing and explains $\text{Fix}(U) = \text{Fix}(L_A) \cap \text{Fix}(L_B)$
- **$\Rightarrow$ C150 is INTERACTING BUT NOT CHAOTIC — the two are compatible**

Level statistics against generated references



Left: $\langle \tilde{r} \rangle$ per symmetry block with the reference ensembles as bands. Right: the unfolded spacing distribution.

Independent check: the HSF Julia eigendata (rule 6 = 150)

N	states	sectors	#distinct	3^m law	sizes	dim Fix	deg. states
8	256	5	81	81	✓	✓	0.43
9	512	6	121	121	✓	✓	0.53
10	1,024	6	243	243	✓	✓	0.56
11	2,048	7	365	365	✓	✓	0.67
12	4,096	7	729	729	✓	✓	0.66
13	8,192	8	1,093	1,093	✓	✓	0.73
14	16,384	8	2,187	2,187	✓	✓	0.74

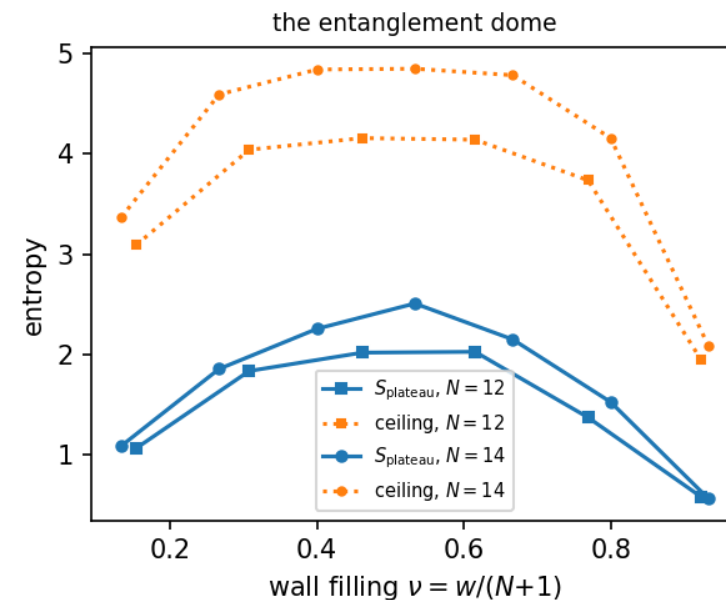
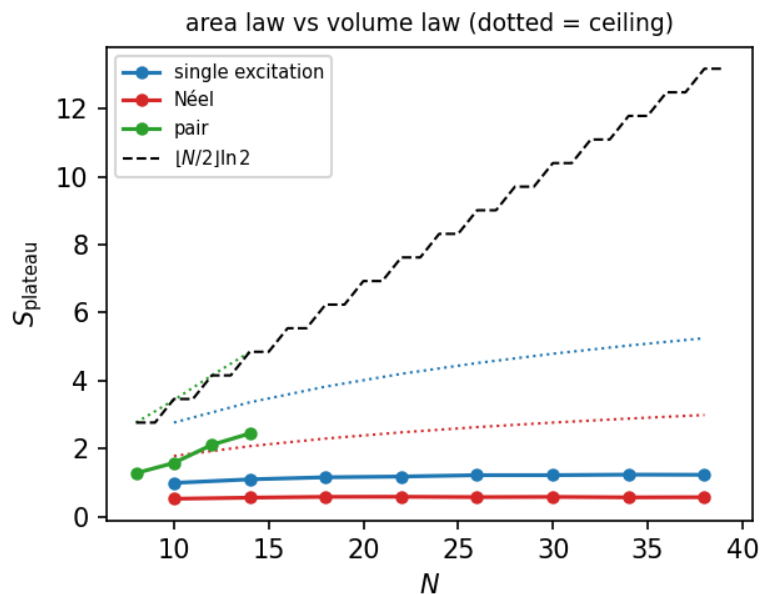
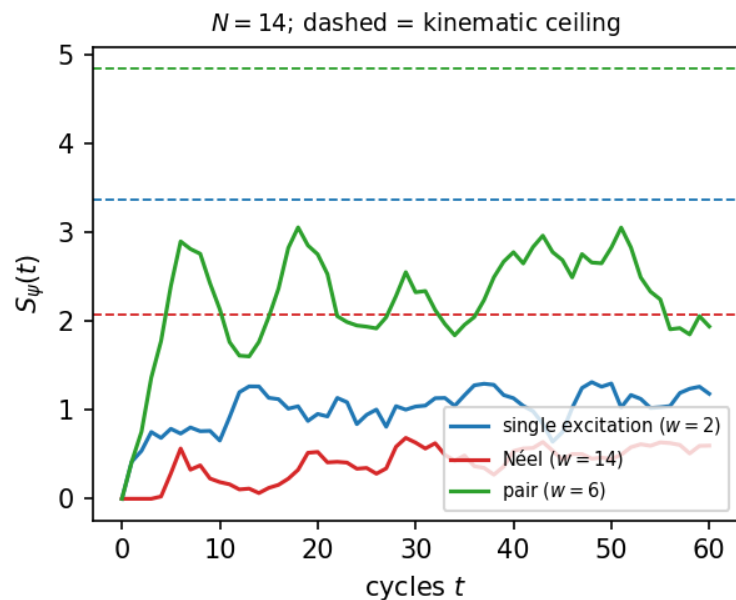
A caveat for the stored entanglement entropies

- **On a degenerate eigenvalue the eigensolver returns an ARBITRARY orthonormal basis**
 - and the half-chain entropy is not constant on that basis
- **Measured**
 - rotating inside one eigenspace moves the stored entropies by up to 0.27 nats
 - the same test on a simple eigenvalue gives a spread of exactly 0
- **For this rule that is the typical case, not an edge case**
 - fraction of stored eigenstates in a degenerate eigenspace: 0.43 (N=8) $\rightarrow$ 0.74 (N=14)
 - multiplicities up to 35 within a single sector
- **Not a defect in the HSF code — a property of the spectrum**
 - options: restrict statistics to simple eigenvalues, or use a basis-independent quantity (eigenspace-averaged purity)
 - worth checking which of the other plotted rules share this

Result 5 — why the same rule gives area AND volume law

- **The three Fig. 3a states are in three very differently sized sectors**
 - single excitation $|0\dots010\dots0\rangle$: $w = 2 \rightarrow |S_w| = C(N+1,2)$, POLYNOMIAL
 - Néel $|0101\dots\rangle$: $w = N \rightarrow |S_w| = N+1$, LINEAR
 - pair $|001100\dots\rangle$: $w \approx N/2 \rightarrow |S_w|$ exponential
- **So the premise fails: $C(N+1,w)$ is exponentially large only near $w = (N+1)/2$**
 - the Néel state is a ONE-HOLE state in wall language — a wall on every bond but one, hard-core $\Rightarrow$ JAMMED
- **The exact ceiling is the bipartite Schmidt rank, not $\ln|S_w|$**
 - from $w = w_A + b_{\text{cut}} + w_B$; the LOWER limit matters at high filling
 - verified = the compressed Schmidt dimension for all w , $N = 6\dots14$
 - at $N=26$: single $\ln 92 = 4.5$, Néel $\ln 14 = 2.6$, pair $9.011 = \lfloor N/2 \rfloor \ln 2$ exactly

Area law, area law, volume law — predicted by w alone



Left: $S(t)$ at $N=14$ with the kinematic ceilings. Middle: the plateau vs N — single excitation and Néel are FLAT to $N=38$, the pair state grows. Right: the entanglement dome vs wall filling.

Is the sector enough? — the three answers

- **For the growth TYPE and its N-scaling: YES**
 - via the exact ceiling, not via $|S_w|$ — knowing w already separates $\ln N$ from N
- **For PRODUCT initial states, the sector nearly fixes the value too: $\pm 3\%$**
 - six random basis states of the $N=14$, $w=6$ shell: plateaus 2.26 ... 2.39
 - Fig. 3a uses product states $\Rightarrow$ its three curves are set by three integers w
- **Over ALL states of a shell: NO**
 - the stationary states of the same shell — $C([N/2], w/2) = 35$ at $N=14$, $w=6$ —
 - have $S(t)$ CONSTANT to $1e-15$, at a value ABOVE the product plateau (3.5 vs 2.4)
 - a 35-dimensional family with zero entanglement growth inside a volume-law sector
- **The dome's shape is the ceiling's shape; the dynamics supplies only an $O(1)$ fill factor $\approx 1/2$**
 - the same sub-thermal fraction as the HSF entropies — as the degenerate spectrum predicts

Open items

- **Identify the commutant**
 - the 3^m law forces an exponentially large commutant ($\sum m_\lambda^2$)
 - its generators would prove the law AND make the block-resolved spacing statistics exact rather than a lower bound
- **Derive the two closed forms**
 - $\dim \text{Fix}(U_w) = C([N/2], w/2)$, and a w -resolved version of the 3^m law
- **The partner-shell intertwiner**
 - for odd N the shells w and $N+1-w$ have identical spectra ($1.2e-14$),
 - but the bond particle-hole map does NOT commute with U — unidentified
- **C150 on the ring**
 - kinematics solved; the quantum layer is obc0 only. The 2-to-1 wall map gives the bond frame a Z_2 gauge redundancy
- **Frontier**
 - $N = 33$ needs a 64-bit label array (68.7 GB), or a two-pass compact relabelling

Reproduce

- **Closed forms, wall bijection, charge conservation, engine cross-check**
 - `python -m qca_fragmentation.c150 --verify`
- **Frontiers**
 - `python -m qca_fragmentation.run_rule --rule 150 --N 21-22 --bc obc0`
 - `python -m qca_fragmentation.c150 --frontier 21-32 --bc obc0`
- **Quantum layer, level statistics, entanglement, HSF check**
 - `python -m qca_fragmentation.scaling.c150_report --quantum`
 - `python -m qca_fragmentation.quantum.rule150_levels --run`
 - `python -m qca_fragmentation.quantum.rule150_entanglement --run`
 - `python -m qca_fragmentation.quantum.hsf_compare --all`
- **Report and data**
 - `reports/pdf/R8_c150.pdf` (18 pp) · `analytics/c150_*.json`
 - full test suite: 403 tests